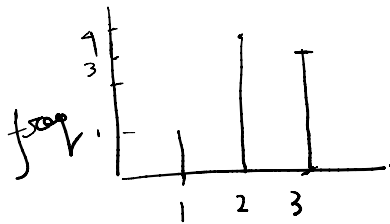


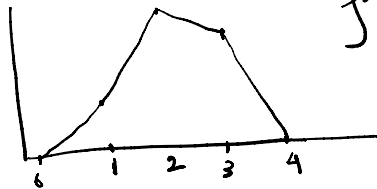
PRST (MCC505)

02 September 2020

09:03



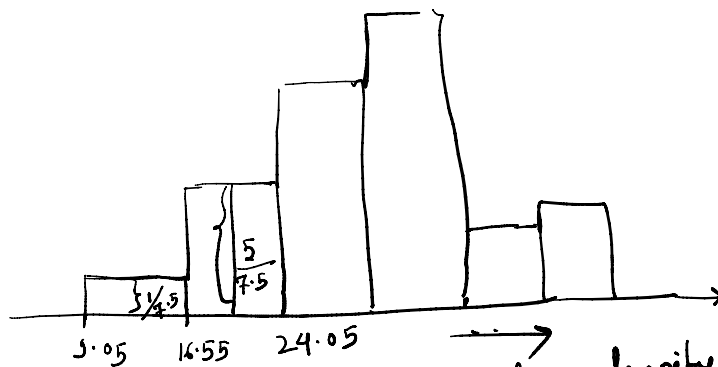
Column diagram



frequency polygon

Class	freq	freq density
9.05 - 16.55	1	$\frac{1}{7.5}$
16.55 - 24.05	5	$\frac{5}{7.5}$
24.05 - 31.55	18	
31.55 - 39.05	21	
39.05 - 46.55	2	
46.55 - 54.05	3	
total n =	50	

$$\text{freq density} = \frac{\text{freq}}{\text{width of the class}}$$



$$\text{relative freq density} = \frac{\text{freq density}}{(\text{total number of observations in the data})}$$

Measure of central tendency:

Mean: $x_1, \dots, x_n$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} (x_1 + x_2 + \dots + x_n)$$

Values freq

x_1	f_1
x_2	f_2
$\vdots$	$\vdots$
x_n	f_n

x_1	f_1
x_2	f_2
$\vdots$	$\vdots$
x_n	f_n

$$\text{total} = \sum_{i=1}^n f_i$$

$$\bar{x} = \frac{(x_1 f_1 + x_2 f_2 + \dots + x_n f_n)}{\sum_{i=1}^n f_i}$$

Class boundaries	freq
9.05 - 16.55	1
16.55 - 24.05	5

class mark for class 1 = 12.8
 " class 2 = 20.3

$$\bar{x} = \frac{12.8 \cdot 1 + 20.3 \cdot 5}{(1+5)}$$

1, 2, 3, 4, mean = 2.5.

1, 2, 3, 100, mean = 24.5

Median: n observations,
 arrange them in increasing
 order.

n: odd, $\frac{n+1}{2}$ th value is the
 median.

1, 2, (3), 4, 5,
 → median

n: even - $\frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$

1, 2, 3, 4, 5, 6

$$\text{median} = \frac{3+4}{2} = 3.5$$

Values freq Cumulative freq

3	5	5
4	9	14
5	11	25 ←
6	4	29
7	1	30

7	1	30
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$$n=30 \quad \frac{n}{2}=15, \quad \frac{n}{2}+1=16$$

$$\text{median} = 5$$

	freq	Cum freq
144.55 — 149.55	1	1
:	:	:
(159.55 — 164.55)	58	(86)
(164.55 — 169.55)	(60)	146
:	:	:
:	:	:
n = 177		

$\frac{n}{2} = 88.5$
 < 88.5
 > 88.5

$$\frac{n}{2} = 88.5$$

$$\text{Median} = 164.55 + \frac{88.5 - 86}{60} \times (169.55 - 164.55)$$